

Model Solution

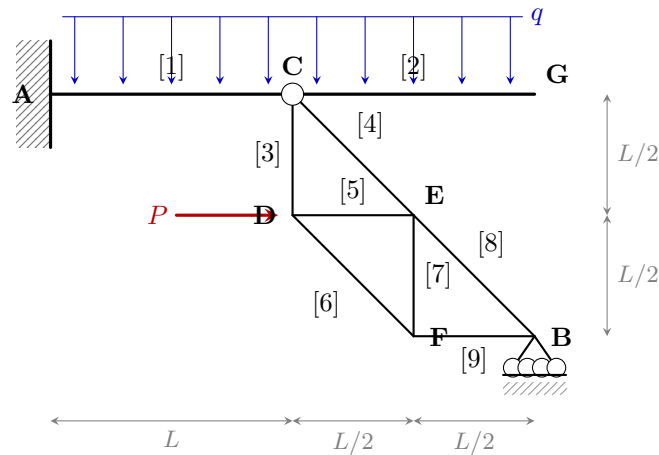
VU Mechanics – LAWI 100515 (PI) SS 2026

2nd Exercise Exam (26.06.2026) – Group A

System values: $L = 1.5 \text{ m}$, $q = 3 \text{ kN/m}$, $P = 15 \text{ kN}$

1. Task

(70 %)



1.1 Static Determinacy

The structure consists of **two rigid bodies**: beam A–C–G (members [1],[2]) and the truss (nodes C,D,E,F,B; members [3]–[9]).

Overall system: 2 bodies $\Rightarrow 2 \times 3 = 6$ equilibrium conditions. Unknowns: support A (fixed, clamped) = 3, support B (roller) = 1, hinge C between beam and truss = 2. Total = 6.

$$n = \underbrace{3}_A + \underbrace{1}_B + \underbrace{2}_C - 2 \cdot 3 = 6 - 6 = 0 \quad \Rightarrow \quad \boxed{n = 0 : \text{statically determinate}}$$

1.2 Support Reactions (general form as functions of q , P , L)

Truss sub-system (joint forces C_x , C_y from beam on truss; B_V at B; P at D):

$$\begin{aligned} \sum M_C = 0 : \quad P \cdot \frac{L}{2} + B_V \cdot L &= 0 & \Rightarrow \quad \boxed{B_V = -\frac{P}{2}} \\ \sum F_x = 0 : \quad C_x + P &= 0 & \Rightarrow \quad C_x = -P \\ \sum F_y = 0 : \quad C_y + B_V &= 0 & \Rightarrow \quad C_y = +\frac{P}{2} \end{aligned}$$

Entire system (resultant distributed load $Q = 2qL$ at $x = L$):

$$\begin{aligned} \sum F_x = 0 : \quad A_H + P &= 0 & \Rightarrow \quad \boxed{A_H = -P} \\ \sum F_y = 0 : \quad A_V + B_V - 2qL &= 0 & \Rightarrow \quad \boxed{A_V = 2qL + \frac{P}{2}} \\ \sum M_A = 0 : \quad M_A - 2qL \cdot L + P \cdot \frac{L}{2} + B_V \cdot 2L &= 0 & \Rightarrow \quad \boxed{M_A = 2qL^2 + \frac{PL}{2}} \end{aligned}$$

1.3 Numerical Values

A_V	A_H	M_A	B_V
$2qL + \frac{P}{2} = \mathbf{16.50 \text{ kN}}$	$-P = \mathbf{-15.00 \text{ kN}}$	$2qL^2 + \frac{PL}{2} = \mathbf{24.75 \text{ kNm}}$	$-\frac{P}{2} = \mathbf{-7.50 \text{ kN}}$

Signs: positive $\uparrow$ for A_V , B_V ; positive $\rightarrow$ for A_H ; CCW for M_A . Negative $\Rightarrow$ actual direction reversed.

1.4 Section Forces in Beams [1] and [2]

At node C the truss applies force $(-C_x, -C_y) = (+P, -\frac{P}{2})$ to the beam (rightward P , downward $\frac{P}{2}$). Running variable x measured from the left end of each beam.

Beam [1] (A $\rightarrow$ C), $0 \leq x \leq L$:

$$N_1(x) = P \quad (\text{constant, tension})$$

$$V_1(x) = \left(2qL + \frac{P}{2}\right) - qx$$

$$M_1(x) = \left(2qL + \frac{P}{2}\right)x - \frac{q}{2}x^2 - \left(2qL^2 + \frac{PL}{2}\right)$$

Beam [2] (C $\rightarrow$ G), $0 \leq x \leq L$:

$$N_2(x) = 0$$

$$V_2(x) = q(L - x)$$

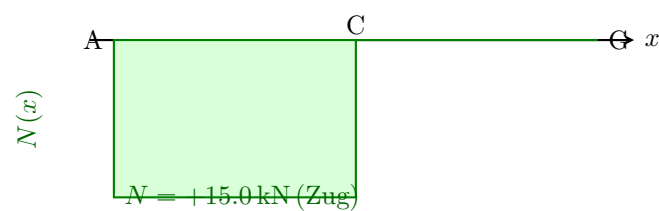
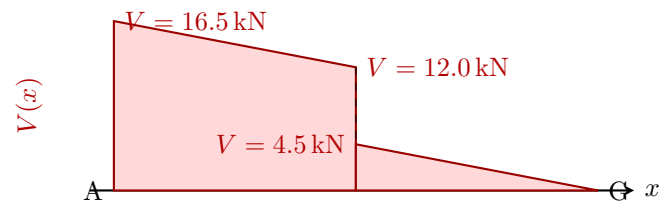
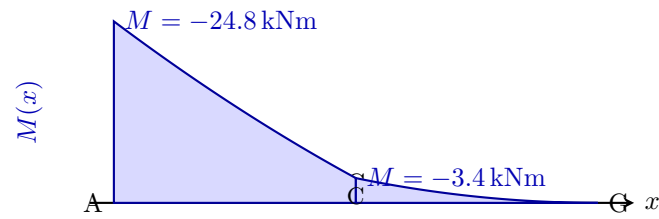
$$M_2(x) = -\frac{q}{2}(L - x)^2$$

At C: shear jump $-\frac{P}{2}$ (vertical truss force); normal-force jump $-P$ (horizontal truss force); moment continuous but with kink.

	Beam start			Beam end		
	M [kNm]	V [kN]	N [kN]	M [kNm]	V [kN]	N [kN]
Beam [1] (A $\rightarrow$ C)	-24.75	16.50	15.00	-3.38	12.00	15.00
Beam [2] (C $\rightarrow$ G)	-3.38	4.50	0.00	-0.00	0.00	0.00

1.5 Distributions $M(x)$, $V(x)$, $N(x)$

Schnittgrößenverläufe (auf der Zugseite; neg. M nach oben)



1.6 Axial Forces S_8 and S_9 – Circular Cut at Node B

At B: member [8] (B→E, direction $\frac{1}{\sqrt{2}}(-1, +1)$), member [9] (B→F, direction $(-1, 0)$), support $B_V = -\frac{P}{2}$.

$$\begin{aligned} \sum F_y = 0 : \quad \frac{S_8}{\sqrt{2}} + B_V = 0 & \Rightarrow \boxed{S_8 = \frac{P}{\sqrt{2}} \approx 10.61 \text{ kN (tension)}} \\ \sum F_x = 0 : \quad -\frac{S_8}{\sqrt{2}} - S_9 = 0 & \Rightarrow \boxed{S_9 = -\frac{P}{2} = -7.50 \text{ kN (compression)}} \end{aligned}$$

1.7 Axial Forces S_4 , S_5 , S_6 – Ritter's Cut

Section through [4],[5],[6]; right part {E,F,B} ($B_V = -\frac{P}{2}$ external; P in left part). Members [4] (C–E) and [6] (D–F): -45° ; member [5] (D–E): horizontal.

S_5 – project all right-part forces onto $\frac{1}{\sqrt{2}}(1, 1)$ (perpendicular to [4] and [6]):

$$-\frac{S_5}{\sqrt{2}} - \frac{P}{2\sqrt{2}} = 0 \Rightarrow \boxed{S_5 = -\frac{P}{2} = -7.50 \text{ kN (compression)}}$$

S_4 – moment sum about D ([5] and [6] pass through D):

$$\frac{S_4}{\sqrt{2}} \cdot \frac{L}{2} - \frac{P}{2} \cdot L = 0 \Rightarrow \boxed{S_4 = \sqrt{2} P = 21.21 \text{ kN (tension)}}$$

S_6 – moment sum about E ([4] and [5] pass through E):

$$-\frac{L}{2} \cdot \frac{S_6}{\sqrt{2}} - \frac{P}{2} \cdot \frac{L}{2} = 0 \Rightarrow \boxed{S_6 = -\frac{P}{\sqrt{2}} = -10.61 \text{ kN (compression)}}$$

S_8	S_9	S_4	S_5	S_6
$\frac{P}{\sqrt{2}} = 10.61$	$-\frac{P}{2} = -7.50$	$\sqrt{2} P = 21.21$	$-\frac{P}{2} = -7.50$	$-\frac{P}{\sqrt{2}} = -10.61$
[kN]; + = tension, - = compression				

2. Task (30 %)

Given: Simply supported beam (span L) with triangular load (0 at A, q_0 at B), $A_V = \frac{1}{6}q_0L = 2.00 \text{ kN}$, $B_V = \frac{1}{3}q_0L = 4.00 \text{ kN}$, and

$$M(x) = -\frac{q_0}{6L}x^3 + \frac{q_0L}{6}x.$$

Values: $L = 1.5 \text{ m}$, $q_0 = 8 \text{ kN/m}$.

2.1 Shear Force $V(x)$

$$V(x) = \frac{dM}{dx} = \frac{q_0L}{6} - \frac{q_0}{2L}x^2$$

Check: $V(0) = \frac{1}{6}q_0L = A_V \checkmark$; $V(L) = -\frac{1}{3}q_0L = -B_V \checkmark$.

2.2 Location of Maximum Moment ($V = 0$)

$$\frac{q_0L}{6} = \frac{q_0}{2L}x^2 \Rightarrow x_{\max} = \frac{L}{\sqrt{3}} \approx 0.8660 \text{ m}$$

2.3 Maximum Moment

$$M_{\max} = \frac{q_0L^2}{9\sqrt{3}} = \frac{\sqrt{3}}{27}q_0L^2 \approx 1.1547 \text{ kNm}$$

2.4 Distributions